Fluctuations of convex hulls of multiple random walks

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Let $Z, Z_1, Z_2, \ldots \in \mathbb{R}^2$ be independent random vectors with the same distribution.

The vectors Z_k will be the increments of a random walk S_n, $n \geq 0$, starting at the origin $\mathbf{0} \in \mathbb{R}^2$, defined by

$$S_0 = \mathbf{0}, \quad \text{and} \quad S_n = \sum_{k=1}^{n} Z_k \quad \text{for } n \geq 1.$$

We are interested in the convex hull

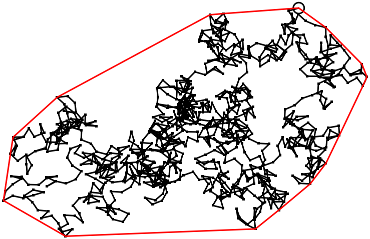
$$\mathcal{H}_n := \operatorname{hull}\{S_0, \ldots, S_n\},$$

that is, the smallest convex set containing the points $S_0, \ldots, S_n$.

For each n, the set $\mathcal{H}_n$ is a (random) nonempty, closed convex set in $\mathbb{R}^2$ whose boundary is a polygon.

We denote

$D_n := \operatorname{diam} \mathcal{H}_n$ the diameter, and
$L_n := \operatorname{perim} \mathcal{H}_n$ its perimeter.



Convex hull of a single path.

Outline

1. Setting
2. Limit theorems for a single random walk
3. Limit theorems for multiple walks
4. Max-type limits for diameter
5. Limit theorems for perimeter

Strong Law of Large Numbers

Assume that $\mathbb{E}[\|Z\|^2] < \infty$. Define

$$\boldsymbol{\mu} := \mathbb{E}[Z], \quad \text{and} \quad \Sigma := \mathbb{E}[(Z - \boldsymbol{\mu})(Z - \boldsymbol{\mu})^\top].$$

The functional form of the strong law of large numbers implies that

$$\lim_{n \to \infty} n^{-1} \mathcal{H}_n = \text{hull}\{\mathbf{0}, \boldsymbol{\mu}\} \quad \text{(a line segment)}, \quad \text{a.s.},$$

where the convergence takes place in the space $\mathcal{K}_0$ of compact convex subsets of $\mathbb{R}^2$ containing $\mathbf{0}$, equipped with the Hausdorff metric ρ_{H}.

Hence, by the continuous mapping theorem,

$$\lim_{n \to \infty} n^{-1} D_n = \|\boldsymbol{\mu}\| \quad \text{and} \quad \lim_{n \to \infty} n^{-1} L_n = 2\|\boldsymbol{\mu}\|, \quad \text{a.s.}$$

$\rightarrow$ McRedmond–Wallace (2018) and McRedmond–Wade (2018).

The Zero Drift Case

When $\boldsymbol{\mu} = \mathbf{0}$, the strong law of large numbers only states that

$$n^{-1}\mathcal{H}_n \to \{\mathbf{0}\}, \quad \text{a.s.}$$

From Donsker's theorem and by continuity, we obtain:

Assume that $\mu = \mathbf{0}$. Let $W : [0,1] \to \mathbb{R}^2$ be a trajectory of standard Brownian motion. Then

$$n^{-1/2}\mathcal{H}_n \xrightarrow[n\to\infty]{\text{dist.}} \Sigma^{1/2}\,\text{hull}\,W[0,1].$$

$\to$ Wade–Xu (2015); Lo–McRedmond–Wallace (2018)

Consequently (assuming that Σ is the identity matrix):

$n^{-1/2}D_n \xrightarrow[n\to\infty]{\text{dist.}} \text{diam}(\text{hull}\,W[0,1])$, and $n^{-1/2}L_n \xrightarrow[n\to\infty]{\text{dist.}} \text{perim}(\text{hull}\,W[0,1])$.

We obtain non-Gaussian limits (positive random variables).

The expected perimeter of the convex hull of Brownian motion is $\mathbb{E}\,\text{perim}(\text{hull}\,W[0,1]) = 2\sqrt{2\pi}$ (Letac–Takács, 1980). On the other hand, the expected diameter is still unknown (available are bounds due to McRedmond–Xu, 2017, and C.–Panzo–Šebek, 2025).

Nonzero Drift – Gaussian Limits

Assume that $\boldsymbol{\mu} \neq \mathbf{0}$. Define

$$\widehat{\boldsymbol{\mu}} := \boldsymbol{\mu}/\|\boldsymbol{\mu}\|, \quad \text{and} \quad \sigma_\mu^2 := \mathbb{V}\mathrm{ar}(\widehat{\boldsymbol{\mu}}^\top Z) = \widehat{\boldsymbol{\mu}}^\top \Sigma \widehat{\boldsymbol{\mu}}. \tag{1}$$

Then, for the diameter, we have (McRedmond–Wade, 2018)

$$\frac{D_n - n\|\boldsymbol{\mu}\|}{\sqrt{n}} \xrightarrow[n\to\infty]{\text{dist.}} \mathcal{N}(0, \sigma_\mu^2). \tag{2}$$

For the perimeter, we have (Wade–Xu, 2015)

$$\frac{L_n - 2n\|\boldsymbol{\mu}\|}{\sqrt{n}} \xrightarrow[n\to\infty]{\text{dist.}} \mathcal{N}(0, 4\sigma_\mu^2). \tag{3}$$

If $\sigma_\mu^2 = 0$, then the convergences (2)–(3) only imply that

$$n^{-1/2}(D_n - n\|\boldsymbol{\mu}\|) \quad \text{and} \quad n^{-1/2}(L_n - 2n\|\boldsymbol{\mu}\|)$$

converge to 0 in probability as $n \to \infty$.
There exist more precise results in this special case.

L^2 Approximation

The convergences in (2)–(3) are proved using martingale difference methods, which allow one to show that when $\boldsymbol{\mu} \neq \mathbf{0}$,

$$\frac{D_n - \widehat{\boldsymbol{\mu}}^\top S_n}{\sqrt{n}} \xrightarrow[n\to\infty]{L^2} 0, \tag{4}$$

and

$$\frac{L_n - 2\widehat{\boldsymbol{\mu}}^\top S_n}{\sqrt{n}} \xrightarrow[n\to\infty]{L^2} 0. \tag{5}$$

The convergences (4)–(5), together with the classical central limit theorem for the random walk $\widehat{\boldsymbol{\mu}}^\top S_n$, lead directly to (2)–(3), and even to a joint convergence result for the vector (D_n, L_n).

Moreover, (4)–(5) yield the variance asymptotics:

$$n^{-1}\,\mathbb{V}\mathrm{ar}\, D_n \to \sigma_\mu^2, \quad n^{-1}\,\mathbb{V}\mathrm{ar}\, L_n \to 4\sigma_\mu^2 \quad \text{as } n \to \infty.$$

Geometric Interpretation

In our work, we show that

$$n^{-1/2}(\|S_n\| - \widehat{\boldsymbol{\mu}}^\top S_n) \xrightarrow[n\to\infty]{L^2} 0,$$

Hence, the convergences (4)–(5) imply

$$\frac{D_n - \operatorname{diam} \ell_n}{\sqrt{n}} \xrightarrow[n\to\infty]{L^2} 0, \quad \text{and} \quad \frac{L_n - \operatorname{perim} \ell_n}{\sqrt{n}} \xrightarrow[n\to\infty]{L^2} 0, \tag{6}$$

where $\ell_n := \operatorname{hull}\{\mathbf{0}, S_n\}$ is the line segment connecting the endpoints of the random walk trajectory.

We conclude that the approximation by a line segment is substantially deeper than what follows merely from the SLLN in the case $\boldsymbol{\mu} \neq \mathbf{0}$.

Remark: A simple analysis shows that there is a certain subtlety hidden in the convergence (6).

Indeed, despite (6), the sets ℓ_n and $\mathcal{H}_n$ differ by fluctuations of order $\sqrt{n}$, i.e. $n^{-1/2}\rho_{\mathrm{H}}(\mathcal{H}_n, \ell_n)$ does not converge to zero in probability.

On the other hand, there exists a metric for which this convergence does hold!

Outline

Notation and Assumptions

Consider N independent random walks $S^{(1)}, \dots, S^{(N)}$ in $\mathbb{R}^2$, where

$$S_0^{(k)} = \mathbf{0}, \quad S_n^{(k)} := \sum_{i=1}^{n} Z_{k,i}, \qquad n \geq 1,$$

and for each $k = 1, \dots, N$, the increments $Z_{k,1}, Z_{k,2}, \dots$ are i.i.d.

Main assumption:

(M) Assume that $\sup_{1 \leq k \leq N} \mathbb{E}[\|Z_{k,1}\|^2] < \infty$, and denote

$$\boldsymbol{\mu}_k := \mathbb{E}[Z_{k,1}] \in \mathbb{R}^2, \quad \Sigma_k := \mathbb{E}[(Z_{k,1} - \boldsymbol{\mu}_k)(Z_{k,1} - \boldsymbol{\mu}_k)^\top].$$

Define also

$$\mathcal{C}_{\boldsymbol{\mu}} := \text{hull}\{\mathbf{0}, \boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_N\}.$$

Consider the joint convex hull

$$\mathcal{H}_n := \text{hull}\{S_i^{(k)} : 1 \leq k \leq N, \, 0 \leq i \leq n\},$$

and its diameter and perimeter:

$$D_n := \text{diam}\, \mathcal{H}_n, \quad \text{and} \quad L_n := \text{perim}\, \mathcal{H}_n.$$

General Goal

The significance of the (possibly degenerate) polygon $\mathcal{C}_{\boldsymbol{\mu}}$ defined in (M) is that it determines the first-order shape of the set $\mathcal{H}_n$ through the SLLN:

$$\lim_{n\to\infty} \rho_{\mathrm{H}}(n^{-1}\mathcal{H}_n, \mathcal{C}_{\boldsymbol{\mu}}) = 0, \quad \text{a.s.}$$

$\to$ Ivanković–Kralj–Sandrić–Šebek (2024)=[IKSŠ]

As a consequence, we obtain the strong laws of large numbers:

$$\lim_{n\to\infty} n^{-1} D_n = \operatorname{diam} \mathcal{C}_{\boldsymbol{\mu}}, \quad \lim_{n\to\infty} n^{-1} L_n = \operatorname{perim} \mathcal{C}_{\boldsymbol{\mu}}, \quad \text{a.s.}$$

(!) The form of the fluctuations in this setting is more complicated than for a single random walk, and this is precisely the topic of our work.

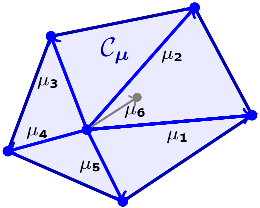
Set approximation

- $\mathcal{J}$ – the set of all $i \in \{0, 1, \ldots, N\}$ such that the point at $\boldsymbol{\mu}_i$. is on the boundary $\partial\mathcal{C}_{\boldsymbol{\mu}}$ of $\mathcal{C}_{\boldsymbol{\mu}}$; we set $\boldsymbol{\mu}_0 := \mathbf{0}$.
- Note that $\mathcal{C}_{\boldsymbol{\mu}} = \text{hull}\{\boldsymbol{\mu}_k : k \in \mathcal{J}\}$.
- $\{\boldsymbol{\mu}_k : k \in \mathcal{J}\} \to$ *boundary drifts*; $S^{(k)}$ with $0 < k \in \mathcal{J} \to$ *extremal walk*.
- Roughly speaking, the extremal walks are the potentially significant ones for the macroscopic asymptotics of $\mathcal{H}_n$.
- Each face of $\mathcal{C}_{\boldsymbol{\mu}}$ is $(\boldsymbol{\mu}_i, \boldsymbol{\mu}_j)$ for some $i, j \in \mathcal{J}$.
- Partition $\mathcal{J}$ into: $\mathcal{J}^0 := \{k \in \mathcal{J} : \boldsymbol{\mu}_k = \mathbf{0}\}$,

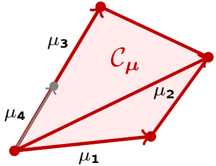
$$\mathcal{J}^{\mathrm{f}} := \{k \in \mathcal{J} : \boldsymbol{\mu}_k \neq \mathbf{0}, (\mathbf{0}, \boldsymbol{\mu}_k) \text{ is contained in a face}\}$$

and

$$\mathcal{J}^{+} := \{k \in \mathcal{J} : \boldsymbol{\mu}_k \neq \mathbf{0}, (\mathbf{0}, \boldsymbol{\mu}_k) \text{ is not contained in a face}\}.$$



$0 \in \mathrm{int}(\mathcal{C}_{\boldsymbol{\mu}})$



$0 \in \partial\mathcal{C}_{\boldsymbol{\mu}}$

- Let

$$\mathcal{G}_n^0 := \left\{S_i^{(k)} : 0 < k \in \mathcal{J}^0,\ i \in \{0, 1, \ldots, n\}\right\};$$
$$\mathcal{G}_n^{\mathsf{f}} := \left\{S_i^{(k)} : k \in \mathcal{J}^{\mathsf{f}},\ i \in \{0, 1, \ldots, n\}\right\};$$
$$\mathcal{G}_n^+ := \left\{S_n^{(k)} : k \in \mathcal{J}^+\right\}.$$

- Note that $\mathcal{G}_n^0$ and $\mathcal{G}_n^{\mathrm{f}}$ contain *whole trajectories* of relevant random walks.
- $\mathcal{G}_n^+$ contains only *endpoints* of non-zero-drift walks.
- For $n \in \mathbb{Z}_+$, define

$$\mathcal{G}_n := \mathrm{hull}\left(\mathcal{G}_n^0 \cup \mathcal{G}_n^{\mathrm{f}} \cup \mathcal{G}_n^+\right) \subseteq \mathcal{H}_n. \tag{7}$$

Theorem.
Suppose that (M) *holds. Then, with* $\mathcal{G}_n$ *defined at* (7),

$$\frac{\rho_{\mathrm{H}}\left(\mathcal{H}_n, \mathcal{G}_n\right)}{\sqrt{n}} \xrightarrow[n\to\infty]{L^2} 0.$$

Outline

Preparation

- We approximated $\mathcal{H}_n$ by $\mathcal{G}_n$, which includes all endpoints of those random walks whose drift is on the boundary of $\mathcal{C}_{\boldsymbol{\mu}}$.
- For the diameter, not all such extremal walks are significant.
- Only those walks that contribute to diametrical chords of $\mathcal{C}_{\boldsymbol{\mu}}$ will contribute to D_n.
- Let

$$J_{\boldsymbol{\mu}} := \{k \in \mathcal{J} : \|\boldsymbol{\mu}_k - \boldsymbol{\mu}_j\| = \operatorname{diam} \mathcal{C}_{\boldsymbol{\mu}} \text{ for some } j \in \mathcal{J}\},$$

  the set of indices that are included in at least one diameter of $\mathcal{C}_{\boldsymbol{\mu}}$.
- Note that every diameter of $\mathcal{C}_{\boldsymbol{\mu}}$ is achieved by a pair of extreme points, so $J_{\boldsymbol{\mu}}$ always contains at least two elements.
- We partition $J_{\boldsymbol{\mu}}$ into $J^0_{\boldsymbol{\mu}} := \{k \in J_{\boldsymbol{\mu}} : \boldsymbol{\mu}_k = \mathbf{0}\}$ and $J^+_{\boldsymbol{\mu}} := \{k \in J_{\boldsymbol{\mu}} : \boldsymbol{\mu}_k \neq \mathbf{0}\}$.

- Let

$$\mathcal{S}_n^0 := \begin{cases} \{ S_i^{(k)} : 0 < k \in J_{\boldsymbol{\mu}}^0,\ i \in \{0,1,\ldots,n\} \}, & \text{if } 0 \notin J_{\boldsymbol{\mu}}; \\ \{\mathbf{0}\} \cup \{ S_i^{(k)} : 0 < k \in J_{\boldsymbol{\mu}}^0,\ i \in \{0,1,\ldots,n\} \}, & \text{if } 0 \in J_{\boldsymbol{\mu}}, \end{cases}$$

  and

$$\mathcal{S}_n^+ := \{ S_n^{(k)} : k \in J_{\boldsymbol{\mu}}^+ \}.$$

  Note that $\mathcal{S}_n^0$ contains *trajectories* of relevant zero-drift walks, while $\mathcal{S}_n^+$ contains only *endpoints* of non-zero-drift walks.
- Define

$$\mathcal{A}_n := \operatorname{hull}\big(\mathcal{S}_n^0 \cup \mathcal{S}_n^+\big). \tag{8}$$

  Note that if there are *no* $0 < k \in J_{\boldsymbol{\mu}}$ for which $\boldsymbol{\mu}_k = \mathbf{0}$, then $\mathcal{S}_n^0$ is either empty (if $0 \notin J_{\boldsymbol{\mu}}$) or equal to $\{\mathbf{0}\}$ (if $0 \in J_{\boldsymbol{\mu}}$).

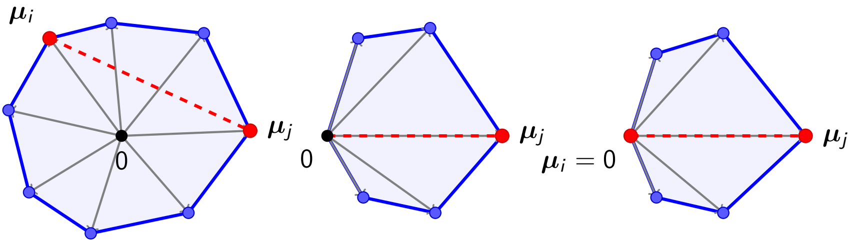
Theorem.

Suppose that (M) *holds. Then With* $\mathcal{A}_n$ *as defined at* (8), *we have*

$$\frac{D_n - \operatorname{diam} \mathcal{A}_n}{\sqrt{n}} \xrightarrow[n\to\infty]{L^2} 0.$$

Random walk approximation

- We can concretely identify an approximation for D_n in terms of the underlying walks (whereas direct computation of $\operatorname{diam} \mathcal{A}_n$, in principle, needs other comparisons that are unlikely to contribute to the diameter).
- Every possible diametrical chord of $\mathcal{C}_{\boldsymbol{\mu}}$ corresponds to a line segment between $\boldsymbol{\mu}_i$ and $\boldsymbol{\mu}_j$ where i, j are two distinct elements from $J_{\boldsymbol{\mu}}$.
- Let
$$\mathcal{D}_{\boldsymbol{\mu}} := \{(i,j) \in J_{\boldsymbol{\mu}}^2 : i < j,\, \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\| = \operatorname{diam} \mathcal{C}_{\boldsymbol{\mu}}\}.$$
Distinguish those diametrical chords that use two non-zero $\boldsymbol{\mu}$s,
$$\mathcal{D}_{\boldsymbol{\mu}}^{+} := \{(i,j) \in \mathcal{D}_{\boldsymbol{\mu}} : \|\boldsymbol{\mu}_i\| \cdot \|\boldsymbol{\mu}_j\| > 0\}, \tag{9}$$
those that use a single non-zero $\boldsymbol{\mu}_j$ and a zero $\boldsymbol{\mu}_i$ (drifts are ordered),
$$\mathcal{D}_{\boldsymbol{\mu}}^{\times} := \{(i,j) \in \mathcal{D}_{\boldsymbol{\mu}} : \boldsymbol{\mu}_i = \mathbf{0} \neq \boldsymbol{\mu}_j\}, \tag{10}$$
and those that use a single non-zero $\boldsymbol{\mu}_j$ and the origin
$$\mathcal{D}_{\boldsymbol{\mu}}^{\circ} := \{(i,j) \in \mathcal{D}_{\boldsymbol{\mu}} : i = 0,\, \boldsymbol{\mu}_j \neq \mathbf{0}\}. \tag{11}$$



$\mathcal{D}^{+}_{\boldsymbol{\mu}}$: two non-zero drifts $\quad$ $\mathcal{D}^{\circ}_{\boldsymbol{\mu}}$: origin vs non zero $\quad$ $\mathcal{D}^{\times}_{\boldsymbol{\mu}}$: zero drift vs non zero

Theorem.
Using the notation at (9)–(11), *we have*

$$\frac{D_n - \max\left(\max_{(i,j)\in\mathcal{D}^+_{\boldsymbol{\mu}}} \|S_n^{(i,j)}\|, \max_{(0,j)\in\mathcal{D}^\circ_{\boldsymbol{\mu}}} \|S_n^{(j)}\|, \max_{(i,j)\in\mathcal{D}^\times_{\boldsymbol{\mu}}} \max_{0\le k\le n} \|S_n^{(j)} - S_k^{(i)}\| \right)}{\sqrt{n}} \xrightarrow[n\to\infty]{L^2} 0.$$

Distributional limit

- Let $\Sigma_{i,j} := \Sigma_i + \Sigma_j$ and $\widetilde{Z}_{k,1} := Z_{k,1} - \boldsymbol{\mu}_k$; then $\mathbb{E}[(\widetilde{Z}_{i,1} - \widetilde{Z}_{j,1})(\widetilde{Z}_{i,1} - \widetilde{Z}_{j,1})^\top] = \Sigma_{i,j}$.
- We also define, analogously to (1), for $i, j \in \{1, \ldots, N\}$,

$$\sigma^2_{i,j} := \widehat{\boldsymbol{\mu}}^\top_{i,j}\Sigma_{i,j}\widehat{\boldsymbol{\mu}}_{i,j} \quad \text{and} \quad \sigma^2_{0,j} := \sigma^2_j := \widehat{\boldsymbol{\mu}}^\top_j\Sigma_j\widehat{\boldsymbol{\mu}}_j. \tag{12}$$

- We describe the limit distribution for the centred and scaled D_n in terms of a particular multivariate Gaussian collection ζ and a non-Gaussian collection ξ (ζ and ξ will be independent).

- Let
$$\zeta = (\zeta_e : e \in \mathcal{D}^+_{\boldsymbol{\mu}} \cup \mathcal{D}^\circ_{\boldsymbol{\mu}}), \text{ and } \xi = (\xi_e : e \in \mathcal{D}^\times_{\boldsymbol{\mu}}). \tag{13}$$
- The marginals of ζ are, for each $e \in \mathcal{D}^+_{\boldsymbol{\mu}} \cup \mathcal{D}^\circ_{\boldsymbol{\mu}}$ and with σ_e^2 given by (12),
$$\zeta_e \sim \mathcal{N}\big(0, 4\sigma_\mu^2 e\big). \tag{14}$$
- ζ_{e_1} and ζ_{e_2} are independent unless e_1 and e_2 share a common non-zero index.
- For such cases, we introduce the notation $e_1 \wedge e_2$ for the common index, and $e_1 \oplus e_2$ to be equal to $+1$ if the common index is in the same position in e_1 and e_2, and -1 otherwise. Then we have
$$\mathbb{C}\mathrm{ov}(\zeta_{e_1}, \zeta_{e_2}) = (e_1 \oplus e_2)\widehat{\boldsymbol{\mu}}_{e_1}^\top \Sigma_{e_1 \wedge e_2} \widehat{\boldsymbol{\mu}}_{e_2}. \tag{15}$$
- The distribution of ζ is specified by the variances (14) and covariances (15).
- The components of ξ are all described in terms of a collection of independent planar Brownian motions $\{(W_t^{(k)})_{t\in\mathbb{R}_+} : 0 < k \in J^0_{\boldsymbol{\mu}}\}$ via
$$\xi_{i,j} := -\inf_{0\le t\le 1} \widehat{\boldsymbol{\mu}}_j^\top \Sigma_i^{1/2} W_t^{(i)}.$$

Theorem.
Suppose that (M) *holds and that* $\boldsymbol{\mu}_k \neq \mathbf{0}$ *for at least one* k. *Then*

$$\frac{D_n - \max\left(\max_{(i,j)\in\mathcal{D}^+_{\boldsymbol{\mu}}} \widehat{\boldsymbol{\mu}}_{i,j}^\top S_n^{(i,j)}, \max_{(\mathbf{0},j)\in\mathcal{D}^\circ_{\boldsymbol{\mu}}} \widehat{\boldsymbol{\mu}}_j^\top S_n^{(j)}, \max_{(i,j)\in\mathcal{D}^\times_{\boldsymbol{\mu}}} \left(\widehat{\boldsymbol{\mu}}_j^\top S_n^{(j)} + \max_{\mathbf{0}\le k\le n} (-\widehat{\boldsymbol{\mu}}_j^\top S_k^{(i)}) \right) \right)}{\sqrt{n}} \xrightarrow[n\to\infty]{L^2} \mathbf{0}.$$

Moreover, with ζ *and* ξ *the random elements from* (13),

$$\frac{D_n - \mathbb{E}[D_n]}{\sqrt{n}} \xrightarrow[n\to\infty]{\text{dist.}} \max\left(\max_{(i,j)\in\mathcal{D}^+_{\boldsymbol{\mu}}} \zeta_{i,j}, \max_{(0,j)\in\mathcal{D}^\circ_{\boldsymbol{\mu}}} \zeta_{0,j}, \max_{(i,j)\in\mathcal{D}^\times_{\boldsymbol{\mu}}} (\zeta_{0,j} + \xi_{i,j}) \right).$$

Application for two walks:

1. (Isosceles drifts I.) If $\|\boldsymbol{\mu}_1\| = \|\boldsymbol{\mu}_2\| > \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\| \geq 0$, then

$$\frac{D_n - \mathbb{E}[D_n]}{\sqrt{n}} \xrightarrow[n\to\infty]{\text{dist.}} \max(\zeta_{0,1}, \zeta_{0,2}),$$

where $\zeta_{0,1}$ and $\zeta_{0,2}$ are independent Gaussians with distribution given in (14).

2. (Isosceles drifts II.) Suppose that $\|\boldsymbol{\mu}_1\| = \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\| > \|\boldsymbol{\mu}_2\| > 0$. Then

$$\frac{D_n - \mathbb{E}[D_n]}{\sqrt{n}} \xrightarrow[n\to\infty]{\text{dist.}} \max(\zeta_{0,1}, \zeta_{1,2}),$$

where $\zeta_{0,1}$ and $\zeta_{1,2}$ are correlated Gaussians with marginals (14) and correlation (15).

3. (Equilateral drifts.) Suppose that $\|\boldsymbol{\mu}_1\| = \|\boldsymbol{\mu}_2\| = \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\| > 0$. Then

$$\frac{D_n - \mathbb{E}[D_n]}{\sqrt{n}} \xrightarrow[n\to\infty]{\text{dist.}} \max(\zeta_{0,1}, \zeta_{0,2}, \zeta_{1,2}),$$

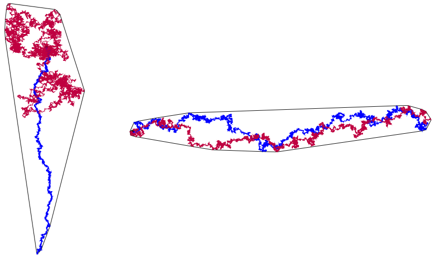
where $\zeta_{0,1}$, $\zeta_{0,2}$ and $\zeta_{1,2}$ are correlated Gaussians with marginals (14) and correlations (15).

4. (Ice-cream.) Suppose that $\|\boldsymbol{\mu}_1\| = 0 < \|\boldsymbol{\mu}_2\|$. Then

$$\frac{D_n - \mathbb{E}[D_n]}{\sqrt{n}} \xrightarrow[n\to\infty]{\text{dist.}} \zeta_{0,2} + \xi_{1,2},$$

where $\zeta_{0,2}$ and $\xi_{1,2}$ are independent; $\zeta_{0,2}$ is Gaussian, and $\xi_{1,2}$ is non-Gaussian.

Two configuration schemes for two random walks



Left: One random walk with zero drift and one with nonzero drift: both the diameter and the perimeter have non-Gaussian limits depending on functionals of Brownian motion.
Right: Two random walks with the same nonzero drift: the limit of the diameter is the maximum of independent Gaussian random variables; the perimeter has a limit given by a sum of a Gaussian random variable plus a random variable with Rayleigh distribution.

Outline

Cauchy formula for compact and convex $K \subset \mathbb{R}^2$:

$$\mathsf{perim}(K) = \int_0^{2\pi} h_K(\theta)\,\mathrm{d}\theta,$$

where

$$h_K(\theta) := \max_{x \in K} x \cdot \mathbf{e}_\theta, \qquad \mathbf{e}_\theta = (\cos\theta, \sin\theta).$$

Notation

- Let $\mathcal{I} := \{1 \le k \le N : \boldsymbol{\mu}_k \in \mathcal{E}\}$ be the set of indices of random walks with drift vectors that are **extreme points** of $\mathcal{C}_{\boldsymbol{\mu}}$.
- Observe that $\mathcal{I} \neq \mathcal{J}$ in general.
- We further split $\mathcal{I} = \mathcal{I}^+ \cup \mathcal{I}^0$ where

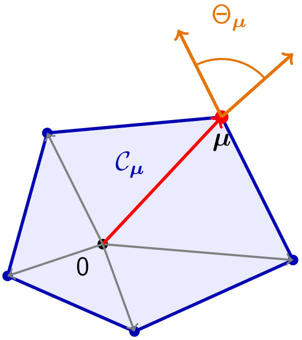
$$\mathcal{I}^+ := \{k \in \mathcal{I} : \boldsymbol{\mu}_k \neq 0\}, \quad \mathcal{I}^0 := \{k \in \mathcal{I} : \boldsymbol{\mu}_k = 0\}.$$

- Define

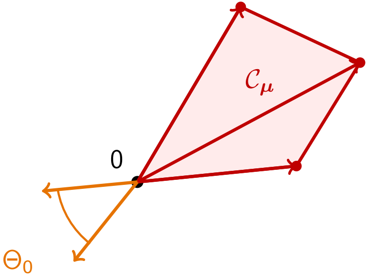
$$\Theta_0 := \Big\{\theta \in [0, 2\pi] : \max_{k \in \mathcal{I}^+} \boldsymbol{\mu}_k \cdot \mathbf{e}_\theta < 0\Big\}.$$

- For each non-zero extreme drift $\boldsymbol{\mu} \in \mathcal{E}$ define

$$\Theta_{\boldsymbol{\mu}} := \mathrm{int}\Big\{\theta \in [0, 2\pi) : \ \boldsymbol{\mu}\cdot\mathbf{e}_\theta = \max_{\boldsymbol{\mu}' \in \mathcal{E}} \boldsymbol{\mu}'\cdot\mathbf{e}_\theta\Big\}, \quad \mathcal{I}_{\boldsymbol{\mu}} := \{k \in \mathcal{I} : \ \boldsymbol{\mu}_k = \boldsymbol{\mu}\}.$$



Directions where $\boldsymbol{\mu}$ dominates.



Directions where 0 supports $\mathcal{C}_\mu$.

Let $(X^{(k)})_{k=1}^{N}$ be independent centred Gaussian vectors

$$X^{(k)} \sim \mathcal{N}(0, \Sigma_k),$$

and let $(B^{(k)})_{k \in \mathcal{I}^{\mathbf{0}}}$ be independent planar Brownian motions with covariance matrices Σ_k.

Theorem.
It holds

$$\frac{L_n - n \,\mathrm{perim}(\mathcal{C}_{\boldsymbol{\mu}})}{\sqrt{n}} \xrightarrow[n\to\infty]{\text{dist.}} Y^{+} + Y^{0},$$

where

$$Y^{+} := \sum_{\boldsymbol{\mu} \in \mathcal{E} \setminus \{0\}} \int_{\Theta_{\boldsymbol{\mu}}} \max_{k \in \mathcal{I}_{\boldsymbol{\mu}}} \left(X^{(k)} \cdot \mathbf{e}_\theta \right) \mathrm{d}\theta,$$

and

$$Y^{0} := \int_{\Theta_{\mathbf{0}}} \max_{k \in \mathcal{I}^{\mathbf{0}}} \sup_{0 \le t \le 1} \left(B^{(k)}(t) \cdot \mathbf{e}_\theta \right) \mathrm{d}\theta.$$

Thank you for your attention!